Product Backlog: MetaForum Study Spot & Group Booking

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Application Overview

A desktop app that simulates booking study spots in a simplified layout of the MetaForum library. Students can see which seats and rooms are free for a chosen time slot, view seat details, and make, view and cancel bookings. Seat occupancy is simulated: other students' bookings are generated randomly when the app starts (no real library data is used).

Goal / outcome: The user finishes with a confirmed booking, shown in a booking confirmation and in the "My Bookings" list.

Advanced Topics

Object Oriented Design Patterns

Git

Core Features

Num	Name	How to Demo	Notes
1	**Login**	Stores student ID (username) and passwords in a .csv file. Program reads .csv file. User can login to existing account or create account.	Cannot login to unexistant account, cannot create existing account.
2	**Start Screen**	Options: Make appointment, Cancel booking. Create account name and password.	
3	**Display floor map**	A simplified floor map is shown, containing rooms drawn as squares.	Simplified layout, not an exact copy of the real building. Rooms are objects.
4	**Select a time slot**	User inputs the start and end time, as well as the date. The appointments in the room "today" will show up in the booking page.	-
5	**Extra info in selector screen**	Room size, utilities.	
6	**Book a room**	Select a room and time, click Book; the appointment is reflected in the booking screen. Appointments and its details are stored in a .csv file.	
7	**Prevent booking conflicts**	Try to book a room at an occupied timeslot or book with an invalid timeslot (end time before start time, duration longer than 2 hours, booking a time that has already passed etc) the booking is refused with an error message.	Back-to-back slots (10–11 and 11–12) are allowed.

Num	Name	How to Demo	Notes
8	**Cancel booking page**	Select a booking, click Cancel and confirm; it disappears from the list. Appointment is also removed from .csv file.	Only can cancel bookings that belong to the currently logged in user.

Extra Features if time allows

Num	Name	How to Demo	Notes
6	**View room details**	Click any seat; a side panel shows its zone, noise level, whether it has a power outlet, and capacity.	Clicking empty space clears the selection.
7	**My Bookings list✓**	Make two bookings; both appear in the "My Bookings" panel, sorted by time.	Map, list and summary update together (Observer pattern).
8	**Show seat availability**	Launch the app; available seats are green and occupied seats are red, with a legend explaining the colours.	Occupancy is randomly pre-filled at startup to simulate other students.
9	**Group room rules**	Try to book a group room; enter a group size. A group larger than the room capacity (or smaller than the minimum, e.g. 3) is rejected with a message.	
10	**Daily booking limit**	Book seats totalling more than 3 hours in one day; the extra booking is refused with a message.	Limit value stored as a constant so it is easy to change.
11	**Save bookings to file**	Make a booking, close the app and reopen it; the booking is still in "My Bookings".	Simple text/CSV file. App starts normally if the file is missing or damaged.